

JULY 01 1989

A theoretical study of cavitation generated by an extracorporeal shock wave lithotripter

Charles C. Church



J. Acoust. Soc. Am. 86, 215–227 (1989)

<https://doi.org/10.1121/1.398328>



LEARN MORE

Advance your science and career as a member of the
Acoustical Society of America

A theoretical study of cavitation generated by an extracorporeal shock wave lithotripter

Charles C. Church^{a)}

Department of Biophysics, School of Medicine and Dentistry, The University of Rochester, Rochester, New York 14642

(Received 8 September 1988; accepted for publication 27 March 1989)

The intense acoustic wave generated at the focus of an extracorporeal shock wave lithotripter is modeled as the impulse response of a parallel RLC circuit. The shock wave consists of a zero rise time positive spike that falls to 0 at $1\ \mu\text{s}$ followed by a negative pressure component $6\ \mu\text{s}$ long with amplitudes scaled to $+1000$ and -160 bars, P^+ and P^- , respectively. This pressure wave drives the Gilmore-Akulichev formulation for bubble dynamics; the zero-order effect of gas diffusion on bubble response is included. The negative pressure component of a 1000-bar shock wave will cause a preexisting bubble in the 1- to $10\text{-}\mu\text{m}$ range to expand to over 100 times its initial size, R_0 , for $250\ \mu\text{s}$, with a peak radius of $\sim 1400\ \mu\text{m}$, then collapse very violently, emitting far UV or soft x-ray photons (black body). Gas diffusion does not appreciably mitigate the amplitude of the pressure wave radiated at the primary collapse, but does significantly reduce the collapse temperature. Diffusion also increases the bubble radius from R_0 up to $40\ \mu\text{m}$ and extends the duration of ringing following the primary collapse, assuming that the bubble does not break up or shed microbubbles. Results are sensitive to P^+ / P^- and to the duration of the negative pressure cycle but not to rise time.

PACS numbers: 43.35.Wa, 43.80.Sh

INTRODUCTION

The use of shock waves to destroy kidney stones and, on an experimental basis, gallstones, has found wide acceptance in hospitals throughout the world. By 1986, Chaussy¹ was able to report that more than 100 000 patients had been successfully treated with the extracorporeal shock wave lithotripter (ESWL) developed by Dornier Medical Systems, GmbH; the current total is over 600 000.² A second class of device, the electrohydraulic lithotripter (EHL), is widely used for the destruction of bladder stones. Both systems offer considerable advantages to patients over surgery, particularly decreased discomfort and a hastened recovery.

Both the ESWL and EHL devices rely on the underwater discharge of an electric spark to produce shock waves. The spark produces a vapor bubble that very quickly expands, forming a shock wave in the liquid in much the same way as an underwater detonation of conventional explosives.³ The devices differ in the positioning of the spark. The EHL spark occurs at the end of a long, flexible cylindrical probe that must be positioned immediately adjacent to the stone, i.e., inside the patient. The ESWL spark occurs at one focus f_1 of an ellipsoidal reflector positioned outside the body; the stone is positioned at the second focus, f_2 . The use of the reflector concentrates approximately 90% of the energy released at f_1 into a focused shock wave at f_2 (Ref. 2). The pressure curves for the two devices also differ. The EHL produces, essentially, a positive pressure pulse only while the ESWL produces a similar positive pulse followed by a long, lower-amplitude negative pressure.⁴ The negative pressure has a profound effect on the computed bubble response, as will be seen.

The mechanisms by which shock waves break up stones are related to "body forces," tensile stresses induced within and at the surfaces of stones, and to cavitation, possibly including both symmetrical and asymmetrical bubble collapses, with the latter producing surface-damaging microjets. The relative importance ascribed to each of these mechanisms varies from author to author,^{5,6} but the physical evidence is available for both.^{4,7,8}

The present work is devoted to describing the results of theoretical computations of the effects of shock waves on free spherical bubbles initially at equilibrium in an infinite liquid. Such bubbles may exist or be produced in the coupling liquid in an ESWL, in the bladder, and, possibly, in the renal cavity. Because bubbles are assumed to retain their spherical shape throughout the duration of their motions, these results may not be directly applicable to the case of a bubble collapsing near the surface of a stone or a fluid-tissue interface. On a qualitative basis, however, these results should still be of some value.

From time to time in the text, comments are made concerning possible physical effects on stones or biological effects on tissue. Such comments are for purposes of illustration only. They are neither completely rigorous nor exhaustive in extent.

1. METHOD

The bubble radius versus time curves presented here are calculated by numerical integration (fourth-order Runge-Kutta) of the Gilmore-Akulichev formulation for bubble dynamics⁹:

$$R \left(1 - \frac{U}{C} \right) \frac{dU}{dt} + \frac{3}{2} \left(1 - \frac{U}{3C} \right) U^2 = \left(1 + \frac{U}{C} \right) H + \frac{U}{C} \left(1 - \frac{U}{C} \right) R \frac{dH}{dR}, \quad (1)$$

^{a)} Current address: National Center for Physical Acoustics, University, MS 38677.

where

$$P = A(\rho/\rho_0)^m - B, \quad (2)$$

$$H = \int_{P_\infty}^{P(R)} \frac{dP}{\rho}, \quad (3)$$

$$C = [C_l^2 + (m-1)H]^{1/2}, \quad (4)$$

$$P(R) = P_g - 2\sigma/R - (4\mu/R)U, \quad (5)$$

and $U = dR/dt$, C is the speed of sound at the bubble wall, H is the enthalpy of the liquid, P and ρ are the time-varying pressure and density of the liquid, respectively, C_l is the infinitesimal speed of sound in the liquid, ρ_0 is the equilibrium liquid density, P_∞ is the pressure at infinity, $P(R)$ is the pressure at the bubble wall, P_g is the pressure of the gas within the bubble, defined below, P_0 is the ambient pressure of the surrounding liquid, σ is the surface tension, and μ is the coefficient of shear viscosity. Following Lastman and Wentzell,¹⁰ we choose $A = C_l^2 \rho / P_0 m$ with $m = 7$ and $B = A - 1$.

The use of Gilmore's equation has two motivations. First, as shown by Vokurka,¹¹ Gilmore's equation is particularly well suited to conditions of high pressure in which the compressibility of the liquid plays an important role. Second, it was desirable to obtain a relatively large quantity of results in a reasonable time. In the formulation used in this study, a polytropic law is used to obtain the pressure inside a bubble; see Eq. (10). A more accurate formalism is available in which heat flow is calculated explicitly,¹² but unfortunately considerable time is involved in obtaining solutions.

The major difference between the present formulation for bubble dynamics and that of Prosperetti *et al.*¹² lies in the inclusion of thermal damping in the latter. Because thermal damping is not an important process in the case of shock wave induced cavitation, Gilmore's equation provides results with reasonable accuracy. Both viscous and acoustic radiation damping are considered here and, as will be seen, the latter is the primary damping mechanism.

Two additional sources of error that usually are not discussed may be of some importance here. The use of a modified Tait equation of state for the liquid, Eq. (2), may introduce considerable error into the calculation when the pressure in the liquid exceeds 10^5 bars. The use of a different equation of state, e.g., the Van der Waals equation at high pressures, would reduce this error although at the expense of increased computing time. On the other hand, the Van der Waals equation requires a calculation of the temperature in the liquid near the surface of the bubble for maximum accuracy, a difficult process.¹² A constant temperature could be used, but would also introduce considerable error. Equation (2) is retained for two reasons: ease of comparison with previous results and the fact that high pressures exist for relatively brief times, thus minimizing this source of error.

A second source of error stems from the large gas compressions predicted for the gas inside bubbles exposed to shock waves. For higher values of P_A , the predicted ratio of bubble radii $R_{\min}/R_0$ is on the order of 0.1 (see Fig. 3), meaning that the volume decreases by a factor of 1000. This implies that the gas density increases by three orders of magnitude, to a value close to that of the surrounding liquid.

Such "condensation" is not consistent with the ideal gas assumed to occupy the bubble; see Eq. (10). On the other hand, because these violent compressions induce very high temperatures in the gas, its true physical state is likely to remain gaseous. The gas will undergo some ionization, but the time scale for the completion of even the first ionization is approximately two orders of magnitude greater than the duration (~ 20 ns) of the high temperatures.¹³ Modeling the material inside the bubble as an ideal gas during this brief time probably does not entail significant error. An exact solution of this problem is beyond the scope of the present work.

The value of $P_\infty = P_0 - P_s(t)$ desired is that measured for an extracorporeal shock wave lithotripter by Coleman *et al.*⁴ The shock wave is modeled as the voltage response of a parallel RLC circuit to a current impulse; the output is redefined in terms of pressure:

$$P_s(t) = 2P_A e^{-\alpha t} \cos(\omega t + \pi/3). \quad (6)$$

Here, P_A is the amplitude of the shock wave, equal to the positive peak pressure P^+ when the rise time is zero, α is the decay constant, $9.1 \times 10^5 \text{ s}^{-1}$, and $\omega = 2\pi f$ is the radial frequency with $f = 83.3 \text{ kHz}$. This choice for f was dictated by the measured duration of the negative pressure component of the shock wave, $6 \mu\text{s}$: $83.3 \text{ kHz} = 1/2 \times 1/6 \mu\text{s}$. The pressure produced by Eq. (6) is shown in Fig. 1. The pressure is very small beyond $t = 7 \mu\text{s}$ and in practice P_A is set to 0 for $t > 7 \mu\text{s}$. The ratio of peak positive to peak negative pressures for this pulse, P^+/P^- , is ~ 6.25 . Thus, if $P^+ = 1000$ bars, $P^- = 160$ bars. The ratio P^+/P^- may be decreased by decreasing α . This has little effect on the shape of the positive pressure phase but increases P^- and, consequently, bubble lifetime, expansion ratio, gas flow, etc.

The rise time from P_0 to $P^+ (= P_A)$ for Eq. (6) is zero. This is done for convenience. In order to investigate the effect of a nonzero rise time, the right-hand side of Eq. (6) is multiplied by the factor $K(1 - e^{-\beta t})$. A 50-ns rise time is achieved by choosing $\beta = 9.21 \times 10^7 \text{ s}^{-1}$. The constant K is chosen as 1.03 in order to match the area under the positive pressure cycle of Eq. (6). While this choice for K would

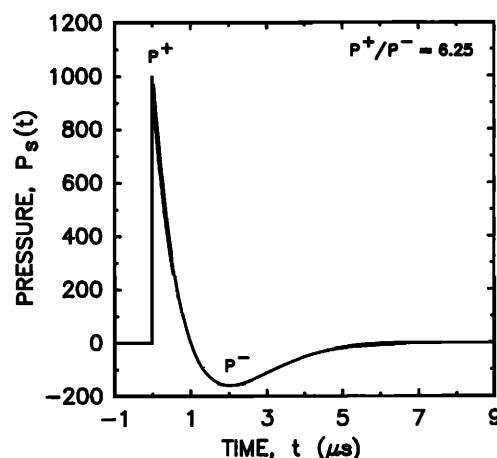


FIG. 1. Modeled form of ESWL pressure waveform as generated by Eq. (6).

increase P^- by 3%, the effect of rise time is explored for positive pulses only. As will be shown, the effect is quite small.

The instantaneous number of moles of gas n in a bubble may be calculated from the zero-order solution to the diffusion equation given by Eller and Flynn,¹⁴ their Eq. (24):

$$n = n_0 - 4(\pi D)^{1/2} \int_0^\tau F(\tau') (\tau - \tau')^{-1/2} d\tau'. \quad (7)$$

The arguments of the integral, a convolution obtained from solution by Laplace transforms, have been interchanged to facilitate the necessary numerical solution. In Eq. (7), D is the diffusion constant of the gas in the liquid, n_0 is the number of moles of gas initially present in the bubble, and

$$\tau = \int_0^\tau R^4(t') dt', \quad (8)$$

$$F(\tau) = C_0(P_g/P_0) - C_i, \quad (9)$$

where

$$P_g = \left(P_0 + \frac{2\sigma}{R_0}\right) \frac{n}{n_0} \left(\frac{R_0}{R}\right)^{3\eta} \left(\frac{R_{0n}}{R_0}\right)^{3(\eta-1)}, \quad (10)$$

R_0 is the initial equilibrium radius of the bubble, R_{0n} is the time-varying equilibrium bubble radius, η is the polytropic exponent of the gas, C_0 is the saturation concentration of the gas in the liquid, and C_i is the initial concentration of gas in the liquid far from the bubble.

Several features of this method for calculating the net flow of gas into a bubble are worth noting. Although Eq. (7) is only the zero-order solution for gas flow, its contribution to the total flow is far greater than that due to all of the higher-order terms combined for short times.^{14,15} For this reason, and because higher-order terms require a considerable increase in computing time, only the zero-order contribution is included. Note, too, that P_g is a function of n . The effect of this is to couple the diffusion equation and the equation of bubble motion, Eq. (1). This requires that the solutions for $R(t)$ and $n(t)$ be obtained simultaneously. Further, n is transcendental, i.e., it is a function of itself. Physically, this simply means that the quantity of gas that flows into a bubble during a particular interval of time is affected by net flow that has previously occurred by virtue of a change in P_g . Computationally, it means that a value for the number of moles of gas equal to $n + (dn/dt)\Delta t$ must be used during the initial calculation of the "new" value of n . Equation (7) passes through three successive solutions, making use of the previous "new" values of n and R_{0n} in order to obtain the same degree of accuracy given by the solution to Eq. (1). However, because both Eqs. (7) and (8) are solved numerically by a simple trapezoidal technique, the ultimate precision on n will be somewhat less than that on R . More precise techniques are available, but the substantial increase in computation time would produce a change in n of only a few percent.

The pressure distribution in the liquid surrounding the bubble is calculated according to the method described by Akulichev.¹⁶ This approximation relies on the method of characteristics used to derive Eq. (1) (see Ref. 17). An invariant of the bubble motion G may be specified at the bubble surface by

$$G = R(H + U^2/2). \quad (11)$$

This value propagates outward along one of the (two) characteristic curves in such a way that it remains unchanged; the other invariant is everywhere constant. The pressure in the liquid is then given as

$$P_r(t) = A \left[\frac{2}{m+1} + \frac{m-1}{m+1} \left(1 + \frac{m+1}{rC_i^2} G \right)^{1/2} \right]^{2m/m-1} - B, \quad (12)$$

where r is the distance from the center of the bubble. The time required for the pressure wave to travel the distance $r - R$ also may be calculated from G , but it is of little concern here. Interested readers may consult Refs. 9, 16, and 17 for a more complete discussion of the pressure distribution or Refs. 17 and 18 for a discussion of the method of characteristics as applied to the problem of bubble dynamics.

The instantaneous force on a bubble due to the direct action of a shock wave, i.e., excluding that produced by the motion of the liquid induced by the shock wave, may be calculated from

$$\hat{F}_s = -V \nabla P_s, \quad (13)$$

where V is the calculated volume of the bubble. The pressure gradient of a shock wave is given by

$$\nabla P_s = \frac{\delta P_s(t)}{\delta x} \hat{i} = \frac{\delta P_s(t)}{\delta t} \frac{dt}{dx} \hat{i} = \frac{-1}{C_{sw}} \frac{\delta P_s(t)}{\delta t} \hat{i}, \quad (14)$$

where $\hat{i}$ is the unit vector and $-C_{sw}$ is the speed of the shock wave in the $\hat{i}$ direction. The time-average force on a bubble is found by numerically integrating Eq. (13) and dividing by the duration of the shock wave. In the results presented here, C_{sw} is taken to be equal to C_l . In reality, the speed of the shock wave may be up to 20% greater than C_l (Ref. 19), and the forces shown may be reduced accordingly. It should be noted that Eq. (13) is not accurate when $P_g/\nabla P_s \gtrsim R_0$ and is, therefore, not used at the leading edge of the shock wave.

The very strong bubble collapse which a shock wave may induce in a free bubble is of concern not only for the mechanical effects that it produces but also for potential chemical effects. In order to estimate such effects, it is necessary to estimate the temperature inside a collapsing bubble. Unfortunately, the use of the polytropic exponent assumption, while convenient, precludes the possibility of calculating temperatures precisely. In this work, temperature will be estimated assuming an adiabatic collapse. Because the maximum bubble radius is attained a relatively long time before the primary collapse and considerable heat flow could occur before that collapse, the maximum bubble radius, R_{max} is not an appropriate choice for this estimate. Instead, the radius $R_{M0.2}$, at which the collapse speed attains a value of Mach 0.2, i.e., $0.2C_l$, is used. This choice is quite arbitrary, of course, but it is reasonable because the length of time between $R_{M0.2}$ and either minimum is quite small, on the order of tens of nanoseconds. It is not likely that this estimate is in error by more than a factor of 2. The form of the equation used to calculate temperature is

$$T = T_0 (R_{M0.2}/R_{min})^{3(\gamma-1)}, \quad (15)$$

where T_0 is the equilibrium temperature of the liquid and γ is the ratio of the specific heats of the gas. A study of the sensi-

tivity of Eq. (15) to choice of numerator, including comparisons to more exact models, is in progress.

The values of the physical constants chosen are applicable to the case of an air bubble in water. Specifically, $C_l = 1.5 \times 10^5$ cm/s, $\rho_0 = 1.0$ g/cm³, $P_0 = 1.01 \times 10^6$ dyn/cm², $D = 2.42 \times 10^{-5}$ cm²/s, $\sigma = 70$ dyn/cm, $\mu = 0.01$ P, $\eta = \gamma = 1.4$, and $T_0 = 293$ K. In the case of glycerol, $C_l = 1.9 \times 10^5$ cm/s, $\rho_0 = 1.26$ g/cm³, $\sigma = 63.4$ dyn/cm, and $\mu = 14.9$ P. The concentration of gas dissolved in the liquid is assumed to be either 0% or 90% of its value at saturation; 90% is approximately the physiological value of gas concentration. Calculations are performed for shock wave pressure amplitudes between 100 and 1000 bars, a range that brackets most measured values.^{1,4,19,20} The radii of bubbles used is 1, 3, 5, 7, and 10 μ m. In most cases, results to be expected for other choices of P_A or R_0 , or both, may be inferred relatively easily from the data presented.

II. RESULTS

A. ESWL-type shock waves

The information presented in this section summarizes a variety of results obtained in this study. In order to simplify the discussion, it will be useful to define a few terms and describe their relationship to a sample $R(t)$ curve. Shown in Fig. 2 is the response of a 3- μ m bubble to a 1000-bar shock wave of the type depicted in Fig. 1. Here and elsewhere in this work, the period T_A is 12 μ s ($= 1/83.3$ kHz) while the amplitude of the shock wave refers to P^+ ; $P^- = P^+/6.25$, except as noted. Also shown in Fig. 2 are several terms that are defined as follows: $R_{\min 1}$ —the bubble radius at the time of the initial compression; $R_{\min 2}$ —the radius at the time of the primary collapse, this is not the second true minimum (cf. Figs. 17 and 18), but is likely to be the second minimum of consequence; $R_{\max}$ —the maximum radius attained by the expanding bubble; t_c —the collapse time, i.e., the duration between the arrival of the shock wave and the primary collapse. Because the general shape of the bubble response remains unchanged over the ranges of R_0 and P_A used here, $R(t)$ curves with $R_0 = 3$ μ m will be used throughout this

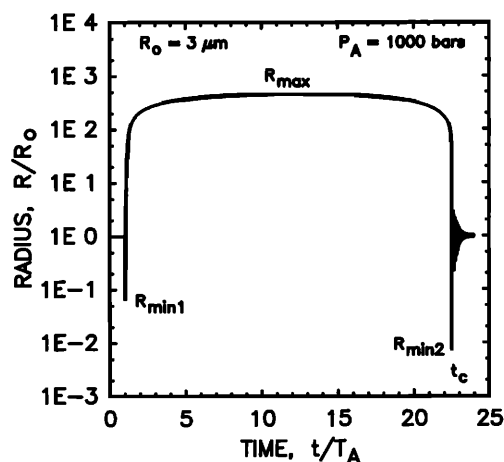


FIG. 2. Bubble response to a 1000-bar ESWL shock calculated without considering gas diffusion, used to define various symbols.

report. No suggestion of bubble resonance phenomena was detected and, given the $R(t)$ curve in Fig. 2, no important resonance effect seems likely to be found in the future.

1. Gas diffusion

Figure 2 was calculated without including the effect gas diffusion, i.e., by setting $n = n_0$ and $R_{on} = R_0$ in Eq. (10). This was done in order to provide a point of comparison between previous calculations of bubble response²¹ (to sinusoidal pressure waves), which were done without considering the effect of diffusion, and the responses given below, e.g., those in Fig. 3. The desirability of including gas diffusion when computing the response of a bubble to a shock wave is obvious from Fig. 2. The bubble expands to a very large size, thus presenting a very large surface area over which diffusion can occur; the gas pressure within the bubble is very low during this expansion, thus providing the pressure gradient needed to drive diffusion; and the expansion persists for a considerable time. This combination of factors permits an enormous amount of gas flow to take place during one "cycle," i.e., during t_c . The effect of this flow is to reduce the violence of the primary collapse and to prolong the subsequent ringing. For the response in Fig. 2, the collapse temperature is calculated to be 8.7×10^6 K and the radiated pressure at $R_{\min 2}$ is 9.95×10^8 bars; these very high estimates provide additional motivation for considering the effects of diffusion. All of the results described below were obtained including the effects of gas diffusion.

The response of a 3- μ m bubble to an ESWL-type shock wave (see Fig. 1) is given in Fig. 3 for pressure amplitudes of 100, 500, and 1000 bars, solid lines. The dashed lines give the net amount of gas inside the bubble normalized to the initial value n_0 . A comparison between the $R(t)$ curve in Fig. 3(c) and that in Fig. 2 highlights the importance of gas diffusion in the case of shock waves. Note that, when additional gas is added to a bubble, the value of $R_{\min 2}$ is increased and the bubble continues to ring for some time following the primary collapse. It is also important to note that the expansion phase of the motion is not affected by gas diffusion. The motion during this phase is controlled almost exclusively by the inertia of the liquid. This is in sharp contrast to the motion following the primary collapse (or to motion due to sinusoidal pressure waves), which exhibits the effect of an interaction between the inertia of the liquid and the pressure of the gas inside the bubble. Although the fact that the relatively large gas flow shown in Fig. 3 may have been expected to produce a noticeable increase in $R_{\max}$, it must be remembered that a stable bubble of size $R_{\max}$ contains approximately 40 000 times more gas than actually enters the bubble during its expansion phase. Thus the additional outward force is quite low and $R_{\max}$ is unaffected. Finally, that the expansion phase is not sensitive to P_g is also demonstrated by the observation that variation in η , see Eq. (10), also has essentially no effect on $R_{\max}$ or t_c (data not shown).

The amount of gas in a bubble at the time of the primary collapse is plotted as a function of pressure amplitude in Fig. 4 for values of $R_0 = 1, 3, 5$, and 10 μ m; the values of n_0 at these radii are 4.1×10^{-16} , 6.9×10^{-15} , 2.8×10^{-14} , and 2.0×10^{-13} moles, respectively. The use of t_c as the temporal

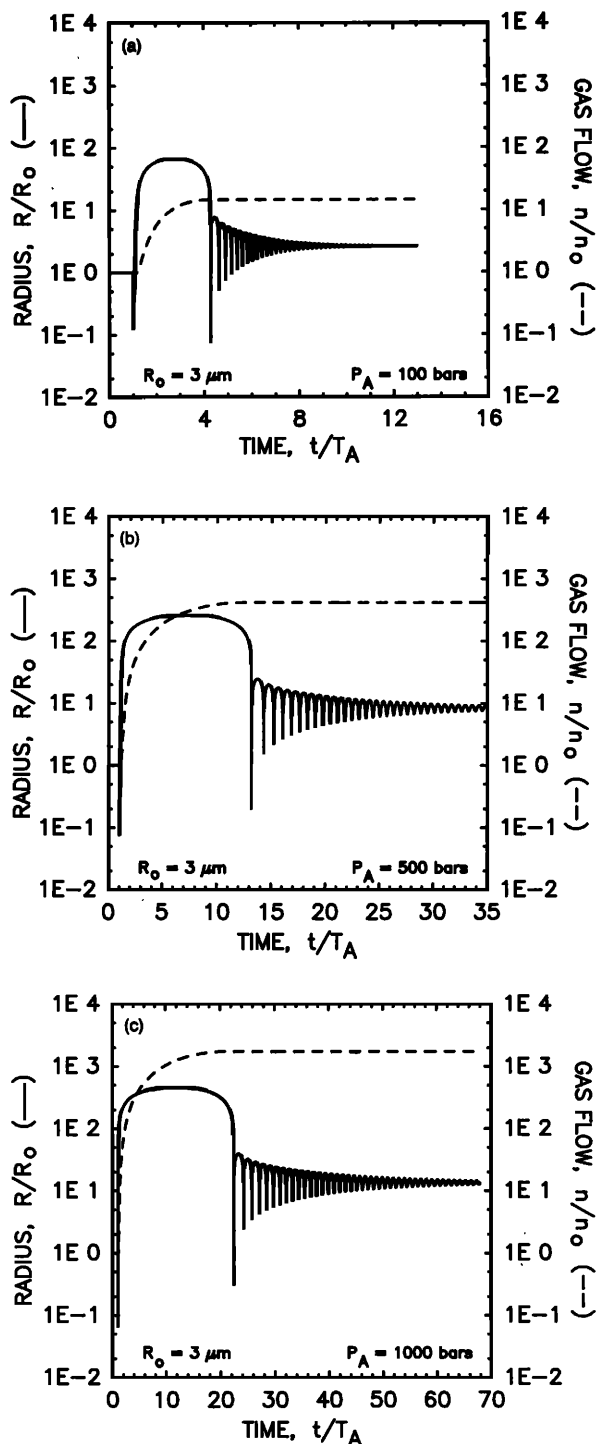


FIG. 3. Responses of a 3- μm bubble, calculated with gas diffusion, to ESWL shocks with amplitudes of (a) 100 bars, (b) 500 bars, and (c) 1000 bars.

marker is a convenient and obvious choice, although slightly less than the maximum amount of gas is present at this time owing to outward diffusion during the collapse. The results indicate that a large volume of gas may enter a bubble if the magnitude of the negative pressure component is high enough. For example, for $R_0 = 1\ \mu\text{m}$ and $P_A = 1000$ bars, $n/n_0 = 28\,775$ at t_c . Although this may seem to be a considerable overestimate, a simple calculation of the thickness of a shell of liquid surrounding the bubble and containing this

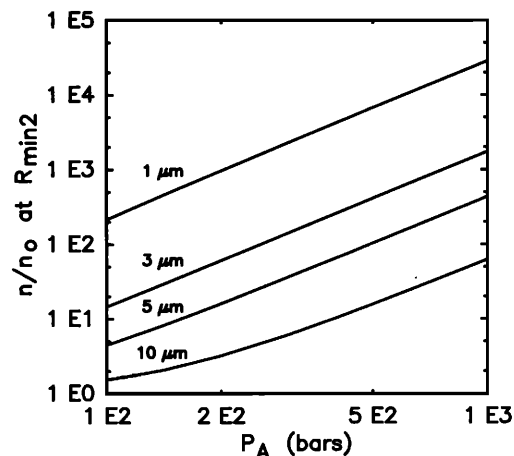


FIG. 4. Number of moles of gas in a bubble at the time of the primary collapse t_c normalized to the initial value n_0 .

much gas shows that the number is quite reasonable. Assuming that the water is 90% saturated with air, this amount of gas is contained in $1.6 \times 10^{-5}\text{-cm}^3$ water. When $R/R_0 = 1000$, the shell is $1.3\ \mu\text{m}$ thick. Since $R/R_0 \geq 1000$ for $190\ \mu\text{s}$ and P_g is very low during this time, the specific value of n/n_0 given here, and by extension the curves in Fig. 4, does not seem unreasonable.

The remarkable amount of gas flow depicted in Fig. 4 has a sizable effect on the equilibrium radius of a bubble. Figure 5 gives the new equilibrium radius of a bubble R_{on} vs P_A at t_c for $R_0 = 1, 5$, and $10\ \mu\text{m}$. Notice that, as P_A increases, bubbles in this range of R_0 tend to grow to about the same size, $40\ \mu\text{m}$; this is also likely to be true for bubbles smaller than $1\ \mu\text{m}$ at $t = 0$. This leads to an unusual prediction, i.e., that any effects due to the primary collapse are independent of the initial radius of the bubble for pressure amplitudes greater than ~ 500 bars. This prediction results from a type of bubble pulsation that is apparently unique to shock waves.^{21,22} Its closest relative is probably the motion of the vapor bubble produced by an underwater explosion.³ The lack of dependence on R_0 at high P_A is reflected in several of the figures presented below.

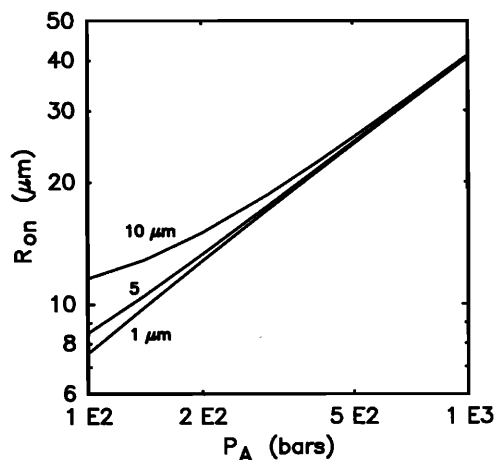


FIG. 5. The equilibrium radius of a bubble at t_c .

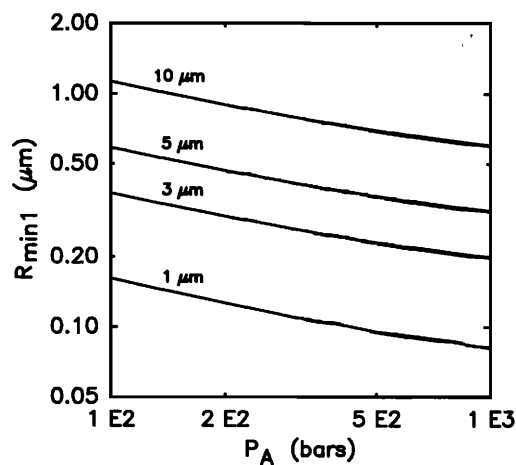


FIG. 6. The radius of a bubble at the initial compression.

Values for the radius of the bubble at three different times, at the initial compression, at the maximum expansion, and at the primary collapse, are given in Figs. 6–8, respectively. The curves for $R_{\min 1}$ show the expected results, namely that smaller bubbles are compressed to smaller radii than are larger bubbles for equivalent pressure amplitudes, and that higher amplitudes produce smaller minima. The curves for $R_{\max}$ in Fig. 7 show the interesting and unexpected result that, for a given pressure amplitude, all bubbles expand to approximately the same size. A similar result was obtained by Noltingk and Neppiras²³ at low frequency and pressure amplitude. This is another indication that it is the inertia of the liquid and not the gas pressure inside the bubble that controls the expansion phase of the bubble motion. The curves for $R_{\min 2}$, Fig. 8, show that, at low pressure amplitudes, the minimum radius is dependent on the initial bubble radius. As P_A increases, however, $R_{\min 2}$ becomes nearly independent of R_0 . This effect is due to the influence of the diffusion of gas into the bubble, as described above.

2. Radiated pressure

The amplitude of the pressure in the liquid surrounding a 3- μm bubble exposed to a 100-bar shock wave is presented

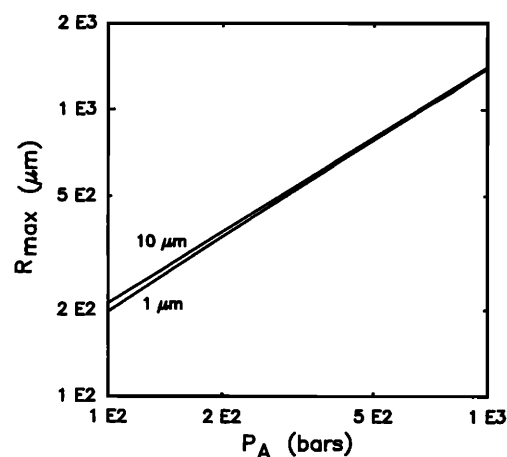


FIG. 7. The maximum radius attained by a bubble.

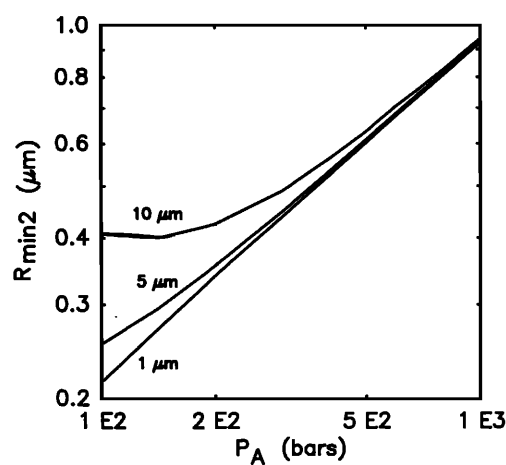


FIG. 8. The radius of a bubble at the primary collapse.

in Fig. 9 for two values of distance from the center of the bubble; the shock wave “arrives” at $t/T_A = 1$. Figure 9(a) shows the pressure at the (moving) bubble surface as a function of time; an offset of 99 bars was added to the calculated pressure in order to accommodate the logarithmic scale. The relatively large negative pressure amplitudes (which are not

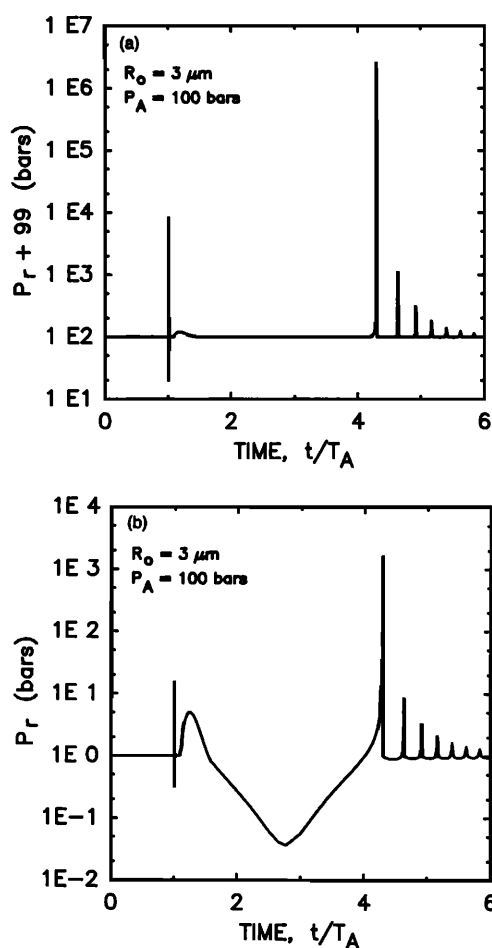


FIG. 9. The pressure in the liquid surrounding a 3- μm bubble exposed to a 100-bar ESWL shock as calculated for a distance from the center of the bubble equal to (a) the moving surface of the bubble $R(t)$ and (b) $R_{\max}$.

resolved on this time scale, see Figs. 17–19) evident near the initial compression are an interesting feature of this curve. These negative pressure pulses probably are not of great importance, at least as far as the breakup of kidney stones is concerned, because they are very brief and closely spaced in time, 10 pulses within 450 ns. Either of the two large positive pressure spikes is likely to be more damaging, particularly the second, due to the primary collapse, which radiates a maximum pressure of 2.6×10^6 bar (5.8 Mbar for $P_A = 1000$ bars). Of course, these calculations are for the case of a spherical bubble in an infinite liquid, but they demonstrate that an enormous amount of energy can be stored by a gas bubble. A simple calculation of the energy deposited in a collapsing bubble by the in-rushing liquid, $E = P_0 dV$, yields from 32.4 to 11 958 ergs over the ranges of R_0 and P_A used here. If this energy is released near an object, either by a radiated shock wave or jet formation, that object is not likely to remain unaffected. The work of Coleman *et al.*⁴ provides striking evidence of the powerful destructive force unleashed by shock wave-induced cavitation.

Figure 9(b) shows the effect of distance traveled on the radiated pressure wave depicted in Fig. 9(a). The pressure is calculated at a distance from the center of the bubble equal to $R_{\max} \sim 200.7 \mu\text{m}$. All amplitudes are considerably reduced, but the primary collapse pressure pulse still exhibits an amplitude of 1700 bars. Since the amplitude of the radiated pressure decays approximately as $1/r$, see Eq. (12), this peak will have an amplitude of 100 bars at a distance of 3.4 mm from the center of the bubble. This observation may have important consequences for measurements of the pressure fields produced by lithotripters.

Plots of the maximum radiated pressure amplitude, i.e., the amplitude at time $= t_c$, as a function of P_A are shown in Fig. 10 for distances from the center of the bubble equal to $R_{\min 2}$ (solid lines) and $R_{\max}$ (dashed lines). There are two lines for each distance, the upper for $R_0 = 1 \mu\text{m}$ and the lower for $R_0 = 10 \mu\text{m}$. Pressures as large as 5.8×10^6 bars are predicted for a 1000-bar shock wave. Such a wave would decay to 2030 bars at $R_{\max} = 1.4$ mm, to 1000 bars at 2.8 mm, and to 284 bars at 1 cm. Detection of cavitation-induced pressure spikes has been reported.^{19,20}

Vogel and Lauterborn²⁴ have measured the amplitude of the pressure wave radiated by a collapsing (really rebounding) laser-produced vapor bubble at a distance of 1 cm from the bubble's center. For $R_{\max} = 1.4$ mm, a value for $P_{r,\max}$ of about 95 bars may be estimated from their data. The corresponding computed value, 284 bars, is high by a factor of 3. Various reasons for this difference may be proposed. For example, the real bubbles probably did not collapse with perfect spherical symmetry, thus reducing the amplitude of the radiated pressure. Similarly, mathematical model does not take proper account of heat conduction, which results in an overestimate for the velocity of the bubble wall during the rebound that produces the outgoing pressure wave²⁵ and thus a larger computed pressure amplitude. Given these considerations, the agreement between the measured and computed values is reasonable.

Experimentally determined values of the yield strengths of renal concretions also are of this order of magnitude (100

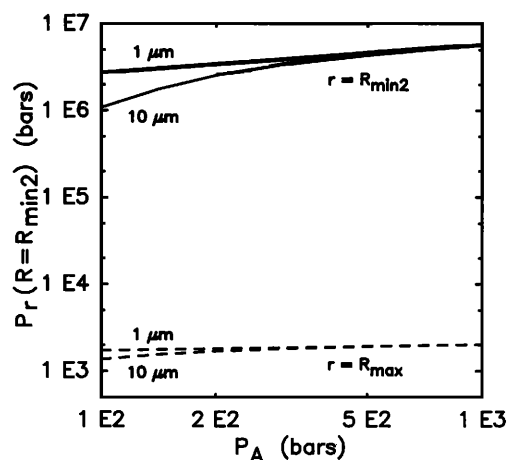


FIG. 10. The pressure in the liquid surrounding a 3- μm bubble at t_c for various P_A as calculated for distances from the center of the bubble equal to $R_{\min 2}$ and $R_{\max}$.

bars). Murata *et al.*²⁶ have measured the compressive strengths of calculi at 19.0 to 176.0 kg/cm² (18.6 to 172.5 bars). Burns *et al.*²⁷ have reported similar values: 24.9 to 76.2 kg/cm² (24.4 to 74.7 bars). Thus it is reasonable to conclude that shock wave induced cavitation may play a significant role in the comminution of kidney stones.

3. Energy balance

As noted above, the amount of energy deposited in a collapsing gas bubble by the in-rushing liquid can be quite high. It is instructive to examine the flow of energy into and, particularly, out of a bubble acted upon by a shock wave. The case of a 3- μm bubble subjected to a 1000-bar shock wave will be considered in detail. The energy absorbed by such a bubble from the impinging shock wave may be computed from $E_{\text{in}} = \int P_s(t) dV$, where the volume V is a function of time. The potential energy stored in the liquid-bubble system at each bubble maximum or minimum, respectively, may be estimated from $E = \int dE = \int [(P_0 - P_g) dV + \sigma dA]$, where dV is the change in volume, and dA the change in surface area, between adjacent extrema. The term $P_g dV$ represents the internal energy of the bubble. Additional contributions to the internal energy are made by the change in the number of moles of gas in the bubble μdn , which is very small, and by heat flow dQ , which is zero by assumption.

The results of these calculations are shown in Table I out to the sixth maximum. The energy $E_{\min 2}$ is deposited in the bubble when it collapses from $R_{\max}$ to $R_{\min 2}$, i.e., the primary collapse. The negative sign indicates the deposition of energy into the bubble by the shock wave. The second column, labeled $\Delta|E|$, gives the amount of energy lost during the motion leading to the preceding extremum. For example, the second entry in this column, 11 540 ergs, is the energy lost by the bubble during its collapse from $R_{\max}$ to $R_{\min 2}$. Column three lists the percentage of the input energy E_{in} which these losses represent. Cumulative losses are shown in column four. Note that 3.1% is lost during the growth to $R_{\max}$ while 96.4% is lost during the primary collapse. Thus

TABLE I. Energy balance for a 3- μm bubble, $P_A = 1000$ bars.

	$E(\text{ergs})$	$\Delta E (\text{ergs})$	% loss	$\Sigma\%$ loss
E_{in}	- 11 976			
$E_{\text{min}2}$	11 600	376	3.140	3.140
$E_{\text{max}2}$	60.435	11 539.565	96.356	99.496
$E_{\text{min}3}$	2.071	58.364	0.487	99.983
$E_{\text{max}3}$	1.280	0.791	0.007	99.990
$E_{\text{min}4}$	0.608	0.672	0.006	99.996
$E_{\text{max}4}$	0.473	0.135	0.001	99.997
$E_{\text{min}5}$	0.306	0.167	0.001	99.998
$E_{\text{max}5}$	0.258	0.048	...	99.998
$E_{\text{min}6}$	0.188	0.070	0.001	99.999
$E_{\text{max}6}$	0.163	0.025	...	99.999

99.5% of all the energy absorbed by the bubble is lost during its first "cycle," i.e., before $t = t_c$. The ringing down that occurs for $t > t_c$, although a significant feature of the $R(t)$ curves, actually represents the loss of very little energy.

The amount of energy in the shock wave radiated by the rebounding bubble following the primary collapse may be computed from

$$E_s = \frac{4\pi r^2}{\rho C_l} \int P_r^2 dt, \quad (16)$$

where r is the distance from the point of measurement, here taken as 1 cm, and the integral is taken over the duration of the radiated wave. For the 3- μm bubble considered above, $E_s \approx 7100$ ergs. This represents 61.5% of the energy lost during the primary collapse or 59.5% of all of the energy initially absorbed by the bubble.

Measurements of some of these quantities have been made by Vogel and Lauterborn²⁴ for nearly spherical bubbles produced by a focused laser beam. They find that on the average a cavitation bubble loses 84% of its energy during its first collapse. Of this, an average of 73% is due to the emission of sound. Thus 61.3% of the energy initially absorbed by the bubble is radiated as an acoustic wave by the collapsing bubble. The agreement between the computed and the measured values is reassuring.

4. Temperature

While the physical effects of cavitation, discussed above in terms of radiated pressure, are likely to be of primary importance when considering the mechanism for the breaking of kidney stones, temperature-mediated chemical changes may produce undesirable side effects in patients or laboratory animals. While the use of the polytropic exponent for determining the gas pressure within the bubble does not permit a precise determination of the temperature of the bubble contents, the use of Eq. (15) does allow reasonable estimates to be made. These estimates are given in Fig. 11 for the initial compression, $R_{\text{min}1}$ (dashed lines), and the primary collapse, $R_{\text{min}2}$ (solid lines), for values of $R_0 = 1$ and 10 μm ; the 1- μm curves are the lower dashed and the upper solid lines.

The temperature curves for the initial compression are important because they tend to confirm the utility of Eq.

(15) in two ways. First, the temperature of a 10- μm bubble is predicted to be greater than that of a 1- μm bubble at a given value of P_A . This is as expected because the small bubble will lose heat more readily than the larger and thus exhibit a lower internal temperature at $R_{\text{min}1}$. Second, the predictions are in the range of a few thousand kelvin. Suslick *et al.*²⁸ have observed temperatures of this magnitude for 20 kHz, cw (sinusoidal) sonications with $P_A = 17.3$ bars. Although it may seem inconsistent to compare a "high"-amplitude shock wave with a "low"-amplitude sinusoid, it should be remembered that the initial compression produced by a shock wave is not preceded by an expansion phase. Bubble expansions tend to lead to more violent, and thus hotter, collapses. This agreement between theory and experiment is taken as support for Eq. (15); a more complete comparison will be presented in a subsequent report.

The upper curves in Fig. 11 show that temperatures up to 10^5 K may be produced by the primary collapse. The production of free radicals by high temperatures is a well-documented outcome of transient cavitation,²⁹ but temperatures of this magnitude may be unique to shock waves. Morgan *et al.*³⁰ have exposed a Fricke chemical (free radical scavenger) dosimeter to 1000 shocks from a Dornier lithotripter and report free radical production equivalent to 1100 rad of ^{60}Co gamma radiation; 1 rad ≈ 100 ergs/cm³ of fluid. In addition to chemical changes, a second effect of such high temperatures would be the emission of high-energy photons. By assuming that the bubble radiates as a black body and calculating the monochromatic emissive power,³¹ it can be shown that the peak in the electromagnetic radiation occurs in the far ultraviolet, $\lambda = 55\text{--}150$ nm. It has long been known that UV light can be damaging to biological tissue.³² The energy flux from a 1- μm bubble exposed to a 1000-bar shock wave is 5.0×10^{15} erg/cm² s, far greater than the solar UV irradiance of 1.6×10^4 erg/cm² s. Of course, the brief duration of the high temperature and the small size of a "collapsed" bubble greatly limit the total energy radiated: flux $\times$ area $\times$ time = 5.4 erg/collapse. The biological consequences of such radiation is unknown.

It is interesting to note that manufacturers of medical

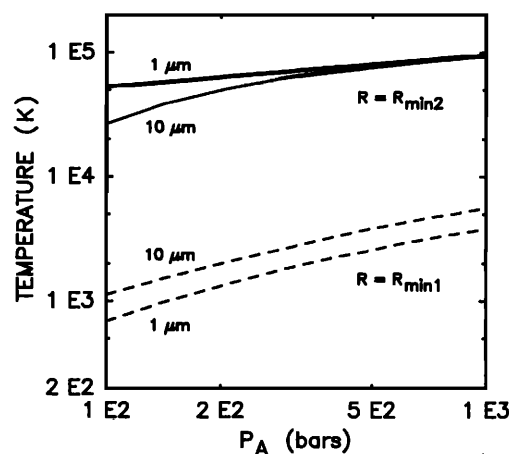


FIG. 11. The temperature inside a bubble at $R_{\text{min}1}$ and $R_{\text{min}2}$ as estimated from Eq. (15).

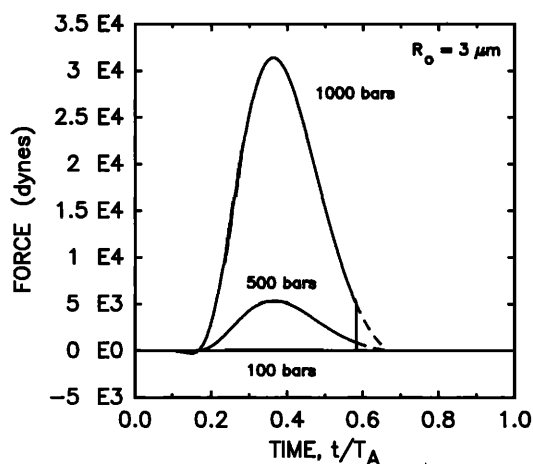


FIG. 12. The force exerted on a 3- μm bubble by ESWL shocks with $P_A = 100, 500$, and 1000 bars.

lithotripters go to great lengths to degas the water which couples the patient, often by immersion, to the source of the shock wave. If the gas concentration of the water were, in fact, significantly reduced while allowing cavitation nuclei to persist, the violence of any cavitation that occurred outside the patient would not be mediated by gas diffusion. The black body radiation from such bubbles would be of a much shorter wavelength, 0.1–3.3 nm for a 1- μm bubble. These are hard x rays, relatively penetrating and biologically dangerous.³² It is reassuring that bubbles are observed to rise up out of the path of the shock wave during patient treatment.³³ Thus, while cavitation nuclei are present in the coupling liquid, there probably is sufficient dissolved gas to render any cavitation bubbles less dangerous.

5. Force on a bubble

The pressure gradient induced by the shock wave in the liquid produces a force on a bubble as described by Eq. (13). The magnitude of the force as a function of time is presented in Fig. 12 with positive values being in the direction of propagation. The solid curves give the force calculated from the truncated form of Eq. (6) actually used in computations of

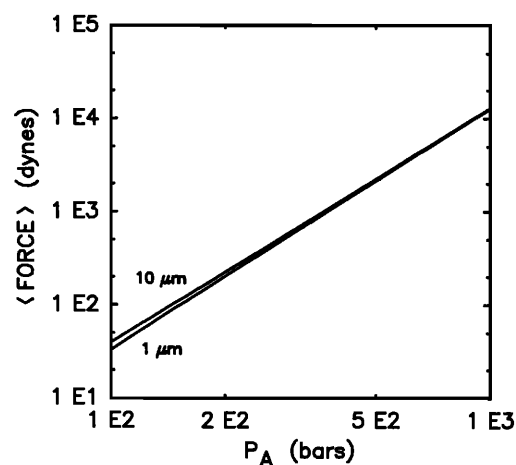


FIG. 13. The time-averaged force on a bubble due to an ESWL shock.

$R(t)$ curves, etc., while the dashed extensions show the smooth return to zero force expected for a real pressure wave. The large peaks in force result from the very large increase in volume produced by the negative pressure cycle of the shock wave. The force at 100 bars is small but nonzero, as may be seen from the time-averaged forces plotted in Fig. 13; averages are over 7 μs . The importance of these forces lies in the fact that the shock wave can push bubbles up against immobile objects, which may then be damaged by subsequent bubble collapses.

An exact calculation of the velocity of a bubble relative to the liquid is rather complicated and of limited interest. A lower bound on the bubble's velocity immediately following the shock wave may be estimated from the time-averaged force on the bubble, the bubble's maximum radius while the shock wave is acting on it (the bubble), and the duration of the shock wave, 7 μs . The bubble experiences an average acceleration equal to the time-averaged force divided by the induced mass of the bubble, i.e., one-half of the mass of the fluid that the bubble displaces. The final velocity is just this acceleration times 7 μs . For a 3- μm bubble exposed to shock waves with $P_A = 100, 500$, and 1000 bars, the calculated velocities are 45.2, 220, and 448 cm/s, respectively. These velocities are quickly reduced by fluid drag, however, and the typical bubble will travel at most a few millimeters.

6. Collapse time

Two features of the timing of the primary collapse merit comment. First, as shown in Fig. 14, t_c is a simple function of P_A and is nearly independent of initial bubble radius. Both observations are true regardless of gas diffusion. It should be possible, therefore, to measure the pressure amplitude of a shock wave by measuring t_c . As the precise location of a cavitation bubble cannot be predetermined, an array of three noncolinear hydrophones may be desirable. Long duration (> 2 ms) pressure traces indicate that the level of noise in such systems is sufficiently low to permit this technique to work.^{19,20}

The second important feature of t_c is that it is not quite independent of R_0 , Fig. 15. Thus, if P_A were known precisely

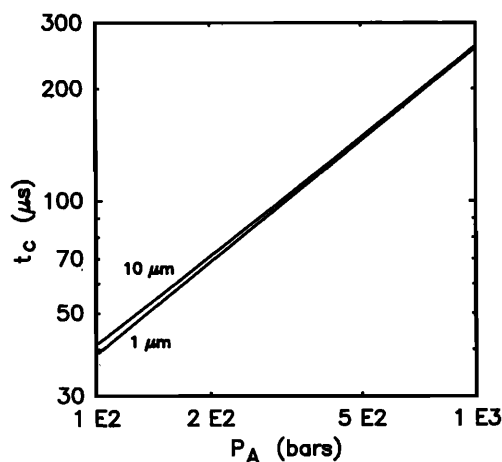


FIG. 14. The time of the primary collapse of a bubble as a function of the pressure amplitude of an ESWL shock.

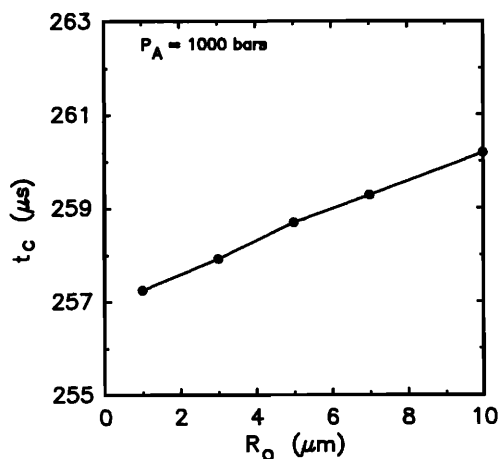


FIG. 15. The collapse time as a function of initial radius for a 1000 bar shock, ●, calculated points.

and if an array of hydrophones were used to reduce the error in the measurement of t_c , then it should be possible to measure R_0 by measuring t_c . Of course, the production of calibrated shock waves by electric sparks would be very difficult, but similar pressure waves may be produced by piezoelectric transducers with little variation in amplitude. Owing to the difficulty of determining P_A and the shallow slope of the curve in Fig. 15, this technique probably would be difficult to implement.

The values of t_c given in Figs. 14 and 15 do not compare favorably with those reported by Hunter *et al.*¹⁹ or Nasr.²⁰ However, an examination of the measured pressure of the shock waves used by these investigators shows that the ratio P^+/P^- is approximately 3.1; for the calculations described above, $P^+/P^- = 6.25$. It is known that the amplitude of the negative pressure cycle has a major impact on the response of a bubble to a sinusoidal pulse.¹⁷ In order to investigate the computational effect of decreasing P^+/P^- to 3.1, the value of α in Eq. (6) was increased to $5.9 \times 10^5 \text{ s}^{-1}$. While this has no effect on P^+ , it does increase the area under the positive pressure cycle of the curve slightly; P_A is again set equal to zero at $t = 7 \mu\text{s}$.

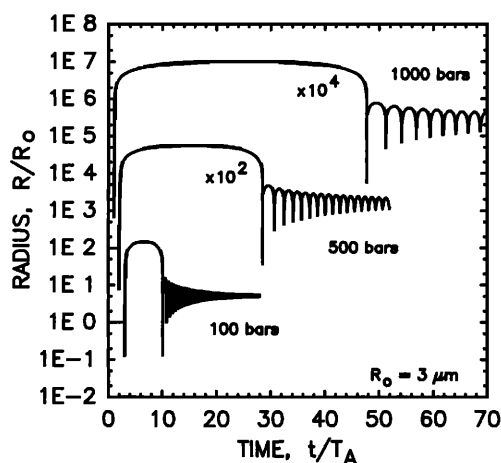


FIG. 16. The response of a 3- μm bubble to ESWL shocks with $P^+/P^- = 3.1$ and $P_A = 100, 500$, and 1000 bars.

Computed responses of a 3- μm bubble to shock waves with this increased negative pressure amplitude are shown in Fig. 16 for $P_A = 100, 500$, and 1000 bars. The calculated values of t_c are 85.1, 318.2, and 560.6 μs , more than double the corresponding values for the curves shown in Figs. 3. An experimental value for t_c of 563 μs taken from Hunter *et al.*¹⁹ for a P_A of ~ 714 bars is in reasonable agreement with these calculations. This value of P_A was measured with a ceramic hydrophone with a response time on the order of 1 μs . Thus the given value was underestimated by an unknown amount. Other sources of difference include uncertainty in the location of the bubble that would result in an increase in distance traveled for the cavitation-generated shock wave and errors in the assumptions upon which the bubble response model is based. Other bubble parameters change to varying extents; R_{0n} increases to $\sim 78.5 \mu\text{m}$ at $P_A = 1000$ bars while the temperature and radiated pressure at the primary collapse increase by 20–30%.

Finally, it is interesting to compare the time $t_{c/2}$ required for the bubble radius to decrease from R_{max} to $R_{\text{min}2}$ to the Rayleigh collapse time for a vapor filled cavity³⁴:

$$T_c = 0.915 R_{\text{max}} [\rho / (P_0 - P_v)]^{1/2}, \quad (17)$$

where P_v is the vapor pressure of the liquid, $2.33 \times 10^4 \text{ dyn/cm}^2$. For a 3- μm bubble exposed to shock waves with amplitudes of 100, 500, and 1000 bars, the computed values of R_{max} are 200.7, 787.7, and 1401.4 μm , respectively. The computed values for $t_{c/2}$ are 18.3, 72.0, and 128.1 μs . The corresponding values for T_c are 18.5, 72.6, and 129.1 μs . The agreement between these pairs of numbers is noteworthy; it may be improved by including P_v in Eq. (5) or by neglecting it in Eq. (17).

B. EHL-type shock waves

As stated previously, the primary difference between ESWL- and EHL-generated shock waves lies in the long negative pressure cycle produced by the ESWL. EHL-type shock waves will be modeled by truncating Eq. (6) at $t = 1 \mu\text{s}$. Although this produces a slightly steeper trailing edge on the pressure curve than is generally observed, it allows a direct comparison with the results of the preceding section.

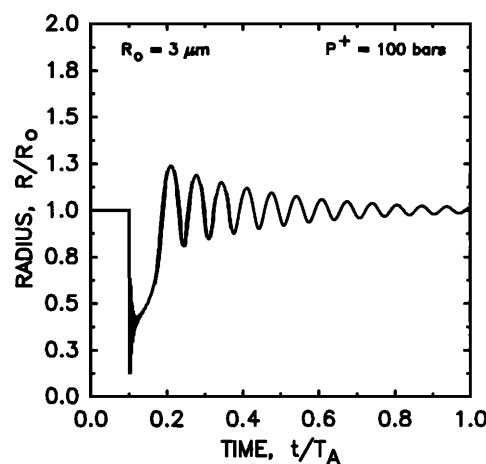


FIG. 17. The response of a 3- μm bubble to a 100-bar EHL shock.

Additionally, a 50-ns rise time will be added to the EHL shock wave rather than to the ESWL shock wave because the small effect is more easily seen.

The response of a 3- μm bubble to a 100-bar shock wave is shown in Fig. 17. The $R(t)$ curve is considerably changed from the curves presented above. There is no long expansion phase, no great influx of gas, and no powerful primary collapse. There is still an impressive "initial" compression; the values of radius, temperature, etc., may be estimated from the data presented above. Following the compression, there are two episodes of ringing, the first in the compressed state and the second following the EHL shock wave. There is a small net force on the bubble in the direction of propagation of the shock wave (calculated with rise time included).

The response of bubbles to EHL shock waves is quite dependent on initial radius. When $R_0 = 10\ \mu\text{m}$, Fig. 18, the response is slower. The ringing of a 3- μm bubble may continue for 20 μs while the larger bubble pulsates for more than 100 μs . Thus the 10- μm bubble absorbs more energy from the shock wave, and releases it more slowly, than does the smaller bubble.

Increasing the pressure amplitude of the shock wave produces a more violent compression, as would be expected. The general shape of the $R(t)$ curve remains unchanged, although the amplitude of the ringing immediately following the compression is reduced while that following the shock wave is increased. The duration of bubble pulsations is also increased as P_A rises.

The effect of rise time on bubble response is shown in Fig. 19 for very brief times. The solid curve gives the response of a 3- μm bubble to a shock wave that reaches 90% of its peak value at 25 ns, and the dashed curve is the zero rise time case. Notice that the two responses are nearly identical, the major difference being seen in the timing of the first minimum. The subsequent responses are indistinguishable.

It is interesting to note that while the responses of gas bubbles to ESWL- and EHL-generated shock waves are quite different, both modalities are effective at breaking stones. Of course, there is a major difference in the positioning of the spark relative to the stone for the two machines. The ESWL spark is outside the body while the EHL probe is

positioned against the targeted concretion. It is possible that the direct action of the vapor bubble itself is of primary importance in stone breakup for EHL devices. If, on the other hand, shock wave-induced cavitation is an important mechanism, then it is the initial compression that must be involved for the EHL. In this case, the effect of the initial compression produced in bubbles by ESWL devices must not be discounted.

C. Tests for cavitation

In attempting to determine the relative importance of the two mechanisms, cavitation or body forces, thought to be responsible for the destruction of kidney stones, it is possible to greatly reduce or eliminate the former while minimally affecting the latter. This is done by altering the physical environment of the stones. For example, a classic test for cavitation effects is increased hydrostatic pressure. Another possibility is to increase the viscosity of the surrounding fluid. If the effects of interest persist after these changes in the system have been made, then cavitation is not responsible. If the effects are reduced or eliminated, then cavitation is probably involved in producing them; such tests cannot offer proof, of course.

Delius *et al.* have reported the results of two such tests on the role of cavitation in the destruction of gallstones *in vitro*.⁸ In the first experiment, an overpressure of 100 bars almost completely eliminated comminution of the stones. This pressure probably destroys most cavitation nuclei initially present in the liquid. Even if bubbles are produced, they will not be particularly active. Calculations of the responses of a 3- μm bubble to a 100-bar overpressure are given in Fig. 20 for $P_A = 500, 700$, and 1000 bars. Notice that the 100-bar overpressure permits cavitation activity when $P_A > 700$ bars. Hunter *et al.*¹⁹ have measured the distribution of P_A versus shock number for a lithotripter similar to that used by Delius *et al.*,⁸ finding that $P_A < 700$ bars for most shocks, but not all. These three pieces of evidence are consistent with a cavitation mechanism.

In the second experiment, stones bathed in glycerol were protected from shock waves while an aqueous solution

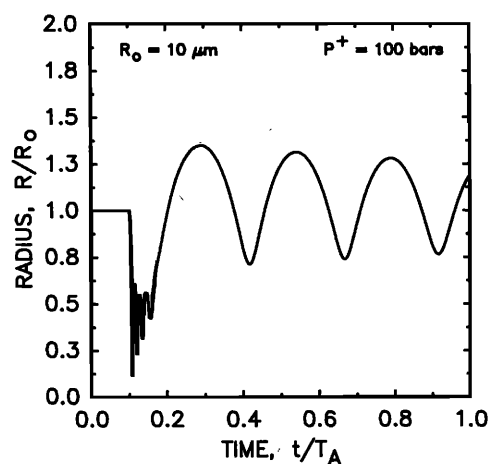


FIG. 18. The response of a 10- μm bubble to a 100-bar EHL shock.

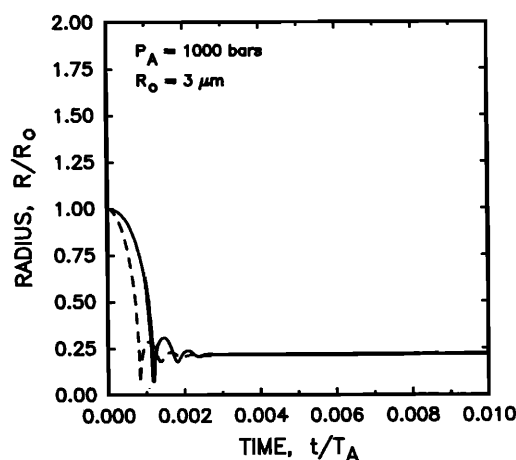


FIG. 19. The response of a 3- μm bubble to a 1000-bar EHL shock including (—) and not including (---) a 50-ns rise time.

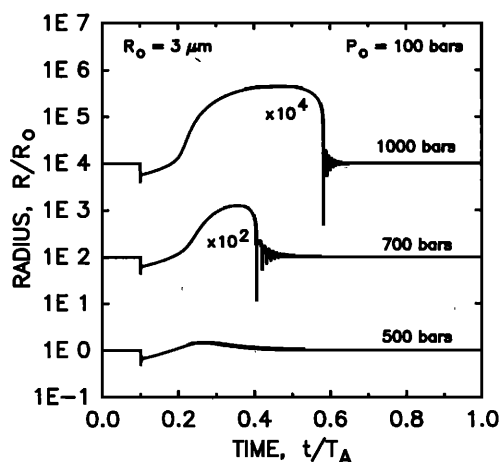


FIG. 20. The response of a 3- μm bubble to ESWL shocks with $P_A = 500$, 700, and 1000 bars for $P_0 = 100$ bars.

of cesium chloride having an acoustical impedance equal to that of glycerol offered no protection.⁸ Calculations of the responses of a 3- μm bubble in glycerol with $P_A = 100$, 500, and 1000 bars are shown in Fig. 21. These results indicate that bubbles in glycerol cannot respond quickly enough to produce violent collapses. There is some question as to whether a shock wave can propagate through a highly viscous liquid and maintain its high pressures. This was tested experimentally by Delius *et al.*⁸ by placing a layer of glycerol, with a thickness equivalent to that used in the original experiment, between the spark source and stones in aqueous cesium chloride. These stones suffered damage almost as severe as did those exposed without an interposed layer of glycerol. The agreement between experiment and theory provides additional support for a cavitation mechanism.

III. DISCUSSION

The computations presented in this paper are for the case of a spherical gas bubble in (usually) water that is acted upon by an ESWL-generated (usually) shock wave. Similar

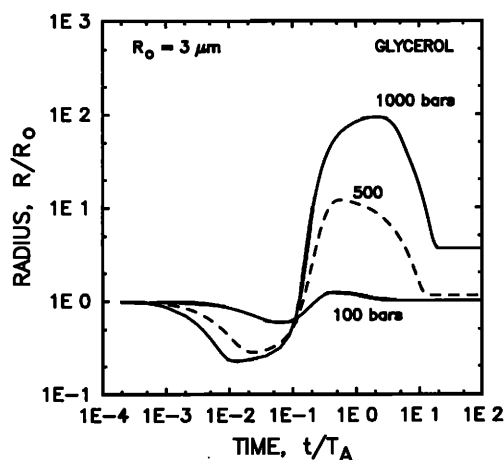


FIG. 21. The response of a 3- μm bubble in glycerol to ESWL shocks with $P_A = 100$, 500, and 1000 bars.

calculations have been performed by Koch *et al.*³⁵ using a model shock wave having a duration approximately twice that employed here. Their results for $R(t)$ curves are generally similar to those presented here, with the exception that R_{max} is 50%–100% larger owing to the longer negative pressure wave.

The present results, as well as those of Koch *et al.*,³⁵ assume that the gas bubble will retain its spherical symmetry throughout its lifetime. Measurements of the rise times for real shock waves yield minimum values of about 30 ns (Ref. 36). Thus the pressure changes on a length scale of about 45 μm . When the shock wave hits a bubble, the pressure difference across its diameter will rise suddenly and the bubble will tend to distort. Because the bubbles considered here are relatively small ($R_0 \leq 10 \mu\text{m}$), and are compressed by a factor of 10 or more almost immediately, the distortion produced in the bubble by this pressure gradient probably has only a relatively minor effect on the bubble's subsequent motion. The agreement between calculations and the various experimental measurements mentioned above provides some justification for this assertion.

If, on the other hand, a bubble collapses near a solid surface, e.g., a kidney stone, then it will quickly lose its spherical symmetry.³⁷ The liquid on the side of the bubble away from the surface will tend to push into and through the gas bubble and strike the solid with sufficient energy to produce considerable local damage.³⁸ Metal foils and plates placed at the second focus of a Dornier ESWL exhibit damage pits characteristic of cavitation bubble jet impacts.^{4,39} Because considerable energy is expended by the bubble in producing a liquid jet, less is available for other effects. As a result, the magnitude of the temperature inside the bubble during the primary collapse, and of the pressure wave radiated immediately afterward, is reduced. Thus bubbles that are actually involved in the destruction of stones probably produce relatively few side effects. Since the jets themselves can be quite damaging, however, cavitation that occurs away from a stone is not a desirable occurrence.

An additional effect of gas diffusion is to increase the probability that a bubble will break up upon collapse. Although a detailed analysis of this process is beyond the scope of the present work, it may be inferred from the work of Birkhoff^{40,41} that the interface of a Raleigh cavity (i.e., one in which the internal pressure is less than ambient and for which the pressure remains constant as the cavity contracts) is stable only for as long as its inward acceleration is increasing in magnitude faster than $1/R^5$ (Ref. 42). This condition is satisfied for any cavity during the initial stages of collapse. For real bubbles the increasing internal gas pressure eventually arrests the inward motion, and thus at some point during the collapse the interface becomes unstable. The additional gas that has diffused into the bubble increases the internal pressure and thus also the radius at which this transition to instability takes place. The stabilizing effect of surface tension is reduced for larger radii.⁴³ Therefore, the sooner that the transition to instability occurs, the greater is the chance that a random perturbation in the shape of the interface will grow large enough to "fracture" the bubble. This tendency of one cavitation bubble to seed others implies that

the role of cavitation in the comminution of stones will increase with the number of shock waves used, provided the seed nuclei do not dissolve in the interim between pulses.

ACKNOWLEDGMENTS

This work was supported by USPHS Grants CA39230-12 and DK39796. The author wishes to thank Professor L. A. Crum for describing his observations of bubble activity in a lithotripter bath and for many helpful comments, Professor H. G. Flynn for valuable discussions on the theory of gas diffusion, and Professor A. Prosperetti for comments on the velocity of a bubble in liquid and energy balance for the bubble system. The support of Dr. M. W. Miller is gratefully acknowledged.

- ¹C. Chaussy, *Extracorporeal Shock Wave Lithotripsy* (Karger, Basel, Switzerland, 1986), 2nd ed., pp. viii, 8–9.
- ²M. Grünwald, H. Koch, and H. Hermeking, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S72 (1988).
- ³R. H. Cole, *Underwater Explosions* (Dover, New York, 1965), pp. 3–13.
- ⁴A. J. Coleman, J. E. Saunders, L. A. Crum, and M. Dyson, *Ultrasound Med. Biol.* **13**, 69–76 (1987).
- ⁵B. Finlayson, M. Nasr, and J. P. Whelan, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S88 (1988).
- ⁶L. A. Crum, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S72 (1988).
- ⁷M. E. Nasr, *Studies of the Fracture Mechanism on Urinary Test Concretions by Lithotripter Devices* (University of Florida, Gainesville, FL, 1982).
- ⁸M. Delius, E. Eizenhoefer, R. Denk, H. Liebich, and W. Brendel, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S88 (1988).
- ⁹E. Cramer, in *Cavitation and Inhomogeneities in Underwater Acoustics*, edited by W. Lauterborn (Springer, New York, 1980), pp. 54–63.
- ¹⁰G. J. Lastman and R. A. Wentzell, *J. Acoust. Soc. Am.* **69**, 638–642 (1981).
- ¹¹K. Vokurka, *Acustica* **59**, 214–219 (1986).
- ¹²A. Prosperetti, L. A. Crum, and K. W. Commander, *J. Acoust. Soc. Am.* **83**, 502–514 (1988).
- ¹³R. W. P. McWhirter, in *Plasma Physics and Nuclear Fusion Research*, edited by R. D. Gill (Academic, London, 1981), pp. 252–256.
- ¹⁴A. Eller and H. G. Flynn, *J. Acoust. Soc. Am.* **37**, 493–501 (1965).
- ¹⁵L. A. Crum and G. M. Hansen, *J. Acoust. Soc. Am.* **72**, 1586–1592 (1982).
- ¹⁶V. A. Akulichev, in *High-Intensity Ultrasonic Fields*, edited by L. D. Rosenberg (Plenum, New York, 1971), pp. 239–259.
- ¹⁷E. J. Ayme, *Transient Cavitation Induced by High Amplitude Diagnostic Ultrasound* (University of Rochester, Rochester, NY, 1988), pp. 41–44.
- ¹⁸F. R. Gilmore, *The Growth or Collapse of a Spherical Bubble in a Viscous Compressible Liquid* (California Institute Technology, Pasadena, CA, 1952), Rep. No. 26-4, pp. 1–40.
- ¹⁹P. T. Hunter, B. Finlayson, R. J. Hirko, W. C. Voreck, R. Walker, S. Walch, and M. Nasr, *J. Urol.* **136**, 733–738 (1986).
- ²⁰M. E. Nasr, *The Evolution in Dynamics of Spark Generated Shock Waves and Their Focusing by Ellipsoidal Reflectors in Lithotripsy* (University of Florida, Gainesville, FL, 1986), pp. 1–163.
- ²¹H. G. Flynn and C. C. Church, *J. Acoust. Soc. Am.* **84**, 985–998 (1988).
- ²²H. G. Flynn, *J. Acoust. Soc. Am.* **72**, 1926–1932 (1982).
- ²³B. E. Noltingk and E. A. Neppiras, *Proc. Phys. Soc. (London)* **B63**, 674–685 (1950).
- ²⁴A. Vogel and W. Lauterborn, *J. Acoust. Soc. Am.* **84**, 719–731 (1988).
- ²⁵A. Prosperetti and A. Lezzi, *J. Fluid Mech.* **168**, 457–478 (1986).
- ²⁶S. Murata, H. Watanabe, T. Takahashi, K. Watanabe, and S. Oinuma, *J. Urol.* **68**, 249–257 (1977).
- ²⁷J. R. Burns, B. E. Shoemaker, J. F. Gauthier, and B. Finlayson, in *Urolithiasis and Related Clinical Research*, edited by P. O. Schwill, L. H. Smith, W. G. Robertson, and W. Vahlensieck (Plenum, New York, 1985), pp. 703–706.
- ²⁸K. S. Suslick, D. A. Hammerton, and R. E. Cline, Jr., *J. Am. Chem. Soc.* **108**, 5641–5642 (1986).
- ²⁹V. Ciaravino, H. G. Flynn, and M. W. Miller, *Ultrasound Med. Biol.* **7**, 159–166 (1981).
- ³⁰T. R. Morgan, V. P. Laudone, W. D. W. Heston, L. Zeitz, and W. R. Fair, *J. Urol.* **139**, 186–189 (1988).
- ³¹D. C. Kelly, *Thermodynamics and Statistical Physics* (Academic, New York, 1973), pp. 350–353.
- ³²D. E. Lea, *Actions of Radiations on Living Cells* (Cambridge U.P., Cambridge, England, 1956), 2nd ed., pp. 3–15.
- ³³L. A. Crum, personal communication (1987).
- ³⁴J. W. Rayleigh, *Philos. Mag.* **34**, 94–98 (1917).
- ³⁵H. Koch, M. Grünwald, and H. Hermeking, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S89 (1988).
- ³⁶A. J. Coleman and J. E. Saunders, *J. Acoust. Soc. Am. Suppl.* **1** **83**, S71 (1988).
- ³⁷Y. Tomita and A. Shima, *J. Fluid Mech.* **169**, 535–564 (1986).
- ³⁸M. Kornfeld and L. Suvorov, *J. Appl. Phys.* **15**, 495–506 (1944).
- ³⁹L. A. Crum, *J. Urol.* **140**, 1587–1590 (1988).
- ⁴⁰G. Birkhoff, *Q. Appl. Math.* **12**, 306–310 (1954).
- ⁴¹G. Birkhoff, *Q. Appl. Math.* **13**, 451–454 (1956).
- ⁴²H. G. Flynn, in *Physical Acoustics, Principles and Methods*, edited by W. P. Mason (Academic, New York, 1964), Vol. 1B, pp. 115–116.
- ⁴³M. S. Plesset, *J. Appl. Phys.* **25**, 96–98 (1954).